AI Structural-Break Detector for Statistical Arbitrage

Point-in-Time Event Intelligence, Multi-Agent Investigation & Decision Control

Complete NLP retrieval, structured extraction, adversarial agent, validation, caching, and deterministic decision-controller specification

This layer begins after the quantitative Model A system has identified and scored a divergence. It reconstructs admissible public evidence at the historical event timestamp and converts that evidence into structured catalyst information. The structured extraction is persisted as the Model B event-feature block and used to obtain the Model B calibrated P(revert); the specialised agents then reason over the same source-grounded evidence plus the Model B quantitative state before validation and deterministic decision control.

Design boundary

Language models are evidence processors and hypothesis generators. They never invent prices, econometric statistics, XGBoost probabilities, or final trading decisions.

Historical integrity

Historical agent runs are restricted to the stored point-in-time evidence corpus. They do not browse the contemporary web during backtests.

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1. Role, system boundary, and end-to-end workflow

The quantitative stack answers “is this a statistically unusual, historically promising divergence?” The language-AI stack answers a different question: “is there contemporaneous public evidence that makes the historical relationship economically unreliable?”

Stage	Input	Output
Event intelligence	event timestamp + both securities + stored SEC filing corpus	Small, admissible evidence set with provenance.
Catalyst extraction	retrieved chunks	Structured event schema for each leg; persisted as Model B event features.
Adversarial analysis	Model B quantitative state + same evidence + extracted events	Independent mean-reversion and structural-break cases.
Validation	agent cases + citations + deterministic inputs	Source/consistency checks and break-risk recommendation.
Controller	pair validity + calibrated P(revert) + break risk + data/liquidity/market-context state	CANDIDATE / REVIEW / NO TRADE / ABSTAIN.

No free-form Decision Agent

The final status is produced by deterministic rules, not by asking an LLM whether to trade. This is essential for reproducibility, bounded behaviour, and historical evaluation.

2. Inputs from earlier layers and typed event state

Input group	Required fields
Event identity	event_id, pair_id, event_timestamp, leg_A security_id, leg_B security_id, divergence_sign.
Econometrics	pair lifecycle status, alpha/beta snapshot, z-score, cointegration statistic/p-value, half-life, beta drift, relationship version.
ML	raw score, calibrated P(revert), probability bucket, top SHAP drivers, model/calibrator/version IDs.
Market context	market/sector/VIX state and realised volatility; all are observable point-in-time covariates.
Implementation	liquidity proxy, daily range/cost proxy, any hard short-side flag available.
Data quality	source freshness, missing features, filing-index status, document retrieval status.

2.1 Typed shared state

LangGraph passes a typed state object rather than an unconstrained chat transcript. Each node may read only the fields required for its mandate and writes schema-validated outputs. This makes agent runs auditable and reduces accidental contamination between roles.

State block	Examples
quant_state	z, P(revert), SHAP drivers, relationship diagnostics, market-context state
evidence_state	retrieved chunk IDs, source timestamps, filing IDs, source leg
catalyst_state	event_type, affected_leg, direction, materiality, relevance, evidence IDs
advocate_state	claim list, evidence links, counterpoints, evidence strength
validation_state	citation validity, contradiction flags, coverage, break-risk recommendation
decision_state	final status, deterministic reason codes, policy version

3. Point-in-time evidence admissibility

The admissibility cutoff is the event timestamp. A source is historical evidence only if its publication/acceptance timestamp is known and is no later than the cutoff. Retrieval time is not a substitute for publication time.

$$\text{admissible}(d, E_i) = \mathbf{1}[\, t_d^{source} \le t_i^{event} \,]$$

Evidence	Primary timestamp	Historical use
SEC 8-K / 10-Q / 10-K	SEC acceptanceDateTime	Core admissible evidence.
SEC amendment	Its own acceptanceDateTime	Admissible only after amendment acceptance.
Optional news	Original publication timestamp	Current/demo enrichment only; excluded from headline historical Model B and Full-System evaluation.
Current web page without historical timestamp	None / retrieval timestamp only	Not admissible for historical evaluation.
Later retrospective article explaining event	After event	Never admissible as pre-event evidence.

Empty evidence is information

If no material filing is found in the lookback window and the source index is complete, the extractor may return event_type=NONE. If the source system itself failed or coverage is unknown, the correct state is UNKNOWN/ABSTAIN, not NONE.

4. SEC filing retrieval policy

SEC EDGAR is the canonical historical text source because it provides reproducible accession numbers and official acceptance timestamps. The retrieval policy selects a small, relevant filing set for both legs before any LLM call.

Document class	Selection rule at event time t
8-K	All 8-K filings accepted during the 30 calendar days ending at t.
10-Q / 10-K	Most recent periodic filing accepted on/before t, normally <=180 days old; extend to <=400 days if necessary.
Amendments	Include only if the amendment itself was accepted before t; retain amendment flag.
EX-99.x exhibits	Include when attached to selected filings, especially earnings releases/guidance.
Foreign forms	Outside MVP domestic-US scope; if later added, apply identical timestamp rules.

4.1 Retrieval query

The evidence selector first filters the SEC filing index by security_id, acceptance timestamp, and permitted form. It then resolves local cached filing bodies/chunks. Network retrieval occurs only for missing selected accession numbers and is cached immediately.

4.2 Evidence window provenance

- Store event_id, security_id, filing_id/accession, document_type, chunk_id, acceptance_datetime_utc, source_url, raw_file_path, and SHA-256 hash.
- Every agent claim cites one or more chunk IDs; UI citations resolve through these IDs.
- The same event/source snapshot must reproduce the same admissible-document set.

5. Document parsing, chunking, and evidence indexing

Stage	Implementation rule
HTML/TXT clean	Remove navigation/boilerplate while preserving headings, item labels, tables-as-text where meaningful, and source metadata.
Section detection	Prefer SEC item/section headings (e.g. Item 2.02, MD&A, Risk Factors) and exhibit boundaries.
Chunking	Keep coherent sections; split oversized text into ~800-1,200-word chunks with ~100-word overlap.
Chunk ID	Stable hash from filing ID + document type + section label + chunk order/content hash.
Index fields	security_id, form, acceptance time, section, document type, chunk text, token estimate, source URL/hash.
Persistence	Write document_chunks Parquet and expose it through DuckDB curated schema.

Raw documents stay immutable

Original SEC files remain in data/raw/sec/filings/... . Parsed chunks are derived objects and can be rebuilt under a parser version. This preserves a defensible audit path from agent claim to original filing.

6. Evidence retrieval and ranking

Retrieval is intentionally two-stage: deterministic admissibility filtering first, then relevance ranking. The language model never receives the entire filing corpus.

Step	Operation
1. Metadata filter	Select chunks for leg A/B whose acceptance time <= event time and whose filings are in the evidence policy.
2. Query construction	Build queries from company name/ticker, known event types, “earnings”, “guidance”, “acquisition”, “regulatory”, “litigation”, “management”, “product”, “financing”, and relationship-change concepts.
3. Core lexical rank	TF-IDF/cosine or BM25-style ranking over the admissible chunk subset; fully local and reproducible.
4. Optional semantic rerank	If time permits, a local embedding model may rerank the top lexical candidates; this is not required for core correctness.
5. Diversity rule	Limit repeated near-duplicate chunks from one exhibit and retain evidence from both legs when available.
6. Evidence pack	Default max 12 chunks total, normally <=6 per leg, ordered with provenance and timestamps.

6.1 Why retrieval is bounded

- Reduces hallucination surface by giving the LLM fewer irrelevant documents.
- Keeps token/API cost predictable.
- Makes human verification feasible because every conclusion points to a small evidence set.
- Prevents a company with a huge filing history from dominating simply because it has more text.

7. Structured catalyst extraction

The Catalyst Extraction Agent converts the evidence pack into machine-readable event objects. It does not decide whether the pair should be traded and does not output a reversion probability.

Field	Allowed values / contract
event_type	EARNINGS, GUIDANCE, M&A, REGULATORY, LITIGATION, MANAGEMENT, PRODUCT, FINANCING, MACRO, SECTOR, NONE, OTHER
affected_leg	A, B, BOTH, UNKNOWN
direction	POSITIVE, NEGATIVE, MIXED, NEUTRAL, UNKNOWN
materiality	LOW, MEDIUM, HIGH, UNKNOWN
relationship_relevance	LOW, MEDIUM, HIGH, UNKNOWN
confidence	0-1 confidence in extraction/classification only; not a trading probability
evidence_ids	Non-empty chunk IDs for every material non-NONE claim
source_timestamp	Latest source timestamp supporting the event; must be <= event cutoff
summary	Short source-bounded explanation; no unsupported external knowledge

7.1 Multiple events

The extractor may return multiple catalyst records. A high-materiality one-sided event must not be diluted by averaging it with unrelated low-materiality items. The downstream validator receives the full list and an explicit dominant-event flag when justified by evidence.

7.2 NONE versus UNKNOWN

State	Meaning
NONE	Evidence retrieval succeeded and no material catalyst was found in the admissible evidence set.
UNKNOWN	Evidence coverage/retrieval/validation is insufficient to make a defensible statement.

8. Multi-agent roles and mandates

Agent	Mandate	Forbidden behaviour
Catalyst Extraction	Classify source-grounded events into the fixed schema.	No trade recommendation; no invented catalyst; no market statistics.
Mean-Reversion Advocate	Build the strongest evidence-based case that the relation remains economically valid.	Cannot ignore adverse evidence; must cite sources/quant state.
Structural-Break Advocate	Build the strongest evidence-based case that new information justifies persistent repricing.	Cannot turn speculation into a fact; must cite sources.
Risk / Validation	Audit both cases, provenance, timestamp validity, contradictions, missing evidence, and mandate compliance.	Cannot change deterministic market/ML inputs or invent citations.
Decision Controller	Deterministic software rules combine validated state.	Not an LLM agent; never free-form.

Adversarial design purpose

The two advocate agents are not votes. They are structured falsification tools: each must construct the strongest defensible case for a competing explanation using the same evidence and quantitative state.

9. LangGraph orchestration and shared state

Node	Reads	Writes
load_event	event_id	quant_state + version IDs
select_evidence	event + source index	admissible document/chunk IDs
rank_evidence	admissible chunks	bounded evidence_pack
extract_catalysts	evidence_pack	catalyst_state
mean_reversion_advocate	quant_state + catalyst_state + evidence	mean_reversion_case
structural_break_advocate	same shared evidence/state	structural_break_case
validate	both cases + sources + deterministic state	validation_state + break-risk recommendation
controller	validated state + policy	decision_state
persist	all validated outputs	DuckDB rows + cache manifest

9.1 Execution topology

After catalyst extraction, the two advocates run in parallel because neither should see the other agent's draft before producing its own case. Their outputs then converge at the Risk / Validation node. This preserves genuine competing hypotheses rather than sequential anchoring.

Bounded execution

Each node has a timeout, retry limit, maximum evidence size, schema validator, and explicit failure state. If a node repeatedly fails, the workflow terminates safely with ABSTAIN rather than hanging the demo.

10. Mean-Reversion Advocate

10.1 Inputs

- Current z-score and direction, relationship validity, half-life, beta stability, recent crossing history.
- Calibrated P(revert) and selected SHAP drivers, clearly marked as model outputs rather than facts about causality.
- Market context/market/sector state.
- Identical catalyst records and evidence pack used by the opposing advocate.

10.2 Required output

Field	Meaning
thesis	Short statement of why the historical relation may remain economically valid.
supporting_claims	List of evidence-backed claims; each has evidence_ids or quant field references.
counterevidence_acknowledged	Material facts that weaken the mean-reversion case.
evidence_strength	LOW/MEDIUM/HIGH based on source support, not rhetorical confidence.
uncertainties	Missing/ambiguous evidence.
source_ids	All cited chunk IDs.

10.3 Appropriate evidence

- Sector-wide movement affecting both firms.
- No material one-sided filings despite complete source coverage.
- Stable rolling pair diagnostics and modest beta drift.
- Prior similar divergences, only if analogue retrieval is explicitly implemented point-in-time; otherwise omit.
- Quantitative score/SHAP as supporting state, never as proof that the structural-break hypothesis is false.

11. Structural-Break Advocate

11.1 Required output

Field	Meaning
thesis	Why current public information may justify a persistent relative repricing.
candidate_break_events	Referenced structured events ranked by materiality/relevance.
economic_link	How each event plausibly changes one leg relative to the other.
counterevidence_acknowledged	Facts that make the break interpretation less certain.
evidence_strength	LOW/MEDIUM/HIGH based on validated support.
uncertainties	Missing context or ambiguous source language.
source_ids	All cited chunk IDs.

11.2 High-value structural-break evidence

- One-sided earnings/guidance shock.
- M&A or divestiture that changes business mix or capital structure.
- Regulatory ruling, litigation, patent/product event, or material financing stress.
- Management change only when the filing/evidence gives a plausible relationship-relevant consequence.
- Business-model change or company-specific exposure change.

Materiality is relative, not merely dramatic

A newsworthy event is not automatically a structural break. The agent must explain why it changes the economics of one leg relative to its historical peer relationship.

12. Risk / Validation Agent

The validator is the control point between generative reasoning and the deterministic decision policy. It verifies source use and consistency; it is not a third investment opinion.

Check	Validation rule
Citation existence	Every cited evidence_id must exist in the evidence pack.
Timestamp	All source timestamps <= event cutoff.
Claim support	Material claims must be directly supported or explicitly marked as inference.
Deterministic consistency	Agents may not alter z, P(revert), pair status, market-context state, or other system values.
Contradictions	Surface conflicts between advocates or among sources; do not resolve by rhetorical preference.
Coverage	Distinguish complete “no event found” from failed/incomplete retrieval.
Mandate	Reject outputs containing unauthorised trade instructions, fabricated prices, or unsupported external facts.
Schema	All required fields/types/enums pass Pydantic validation.

12.1 Validation output

Field	Values
validation_status	PASS / PASS_WITH_WARNINGS / FAIL
source_completeness	COMPLETE / PARTIAL / UNKNOWN
contradiction_flag	true / false
invalid_claim_count	integer
break_risk_recommendation	LOW / MEDIUM / HIGH / UNKNOWN
abstain_recommended	true / false
reason_codes	fixed list of machine-readable flags

13. Break-risk synthesis

Break risk is a structured categorical assessment derived from validated event materiality, one-sidedness, relationship relevance, and the adversarial cases. It is deliberately distinct from the XGBoost reversion probability.

Break risk	Working interpretation
LOW	No validated high/medium relationship-relevant one-sided catalyst; source coverage adequate; structural-break case weak.
MEDIUM	Materiality/relevance is moderate, evidence is conflicting, or a one-sided event is plausible but not decisive.
HIGH	Validated high-materiality, high-relevance one-sided event with strong source support, or multiple consistent break indicators.
UNKNOWN	Evidence coverage/validation insufficient; retrieval failure; unresolved contradictions that prevent defensible classification.

No arithmetic pseudo-probability

Do not convert LLM confidence into a fake “83% break probability”. LOW/MEDIUM/HIGH/UNKNOWN is easier to validate and avoids confusing language-model confidence with calibrated financial probability.

14. Deterministic decision controller

The controller is ordinary deterministic Python. It evaluates hard data/risk gates first, then probability and break-risk policy. Every status includes reason codes and a policy version.

Input class	Examples
Pair validity	ACTIVE/WATCH/RETIRED, current cointegration validity, beta stability.
ML output	Calibrated P(revert), probability bucket, model/calibrator/version, missing-feature flags.
Agent output	Validated break risk, event materiality/relevance, affected leg, source completeness, uncertainty.
Market context	Observable market/sector/VIX stress context.
Implementation	Liquidity/cost/short-side flags.
Data quality	Stale/missing market data, source retrieval failure, invalid agent output.

14.1 Evaluation order

Priority	Gate
1	Critical data/source failure -> ABSTAIN.
2	Pair RETIRED/invalid or hard liquidity/implementation failure -> NO TRADE.
3	Validated HIGH break risk or HIGH-materiality one-sided relationship-relevant catalyst -> NO TRADE.
4	LOW model probability -> NO TRADE.
5	Intermediate probability, MEDIUM break risk, stressed market conditions, soft warning, or conflicting evidence -> REVIEW.
6	Only if all hard gates pass, P(revert) is high and break risk LOW -> CANDIDATE.

15. Decision rules and risk gates

Status	Working default conditions
CANDIDATE	Pair ACTIVE; data/source and liquidity gates pass; calibrated P(revert) >= 0.65; break risk LOW; no HIGH-materiality one-sided relationship-relevant event; no unresolved hard warning.
REVIEW	0.45 <= P(revert) < 0.65, or break risk MEDIUM, or unusual/stressed market conditions, conflicting evidence, partial source coverage that is still interpretable, or soft implementation warning.
NO TRADE	P(revert) < 0.45, pair invalid/retired, HIGH break risk, validated HIGH-materiality one-sided break event, or hard implementation/risk failure.
ABSTAIN	Critical evidence/data unavailable, source timestamps unreliable, agent validation FAIL, missing critical model input, or system cannot make a defensible classification.

Policy thresholds are validation choices

0.45 and 0.65 are working starting values. They must be selected/frozen using training/validation evidence before final-test evaluation. The test set cannot be used to make the policy look better.

15.1 Full-system backtest convention

For automated historical strategy evaluation, only CANDIDATE events enter a simulated position. REVIEW, NO TRADE, and ABSTAIN are treated as withheld opportunities, while their separate counts/reasons are still reported.

16. Abstention and failure handling

Failure	System response
No admissible filing found, index healthy	Extractor may return NONE; not automatically an abstention.
SEC index/download unavailable	ABSTAIN if evidence is required for the full-system decision.
Chunk parser fails for one document	Use remaining evidence only if completeness remains acceptable; otherwise ABSTAIN.
LLM timeout	Retry within limit; on repeated failure -> ABSTAIN.
Malformed JSON/schema	One repair/retry path; repeated failure -> validation FAIL / ABSTAIN.
Invalid citation	Validator removes/rejects unsupported claim; high-impact invalidity -> ABSTAIN.
Agents disagree	Not a failure. Validator records contradiction; usually REVIEW unless evidence clearly supports a hard veto.
SHAP unavailable	Does not invalidate P(revert); agent prompt omits SHAP and controller notes explanation unavailable.

16.1 Why abstention is a product feature

The system is explicitly allowed to decline to classify when evidence is not trustworthy. This makes missing historical text, API failures, and LLM uncertainty visible rather than converting them into silent false confidence.

17. Prompting, hallucination, and prompt-injection controls

Control	Implementation
Role / task / format / constraints	Each agent has a fixed role, bounded task, explicit JSON schema, and forbidden outputs.
Source grounding	Every material textual claim requires an evidence ID from the supplied pack.
Document isolation	Retrieved filing text is delimited as untrusted evidence; instructions inside documents are never followed.
No unrestricted browsing	Historical agents cannot call arbitrary web search.
Deterministic numeric state	Prices/statistics/model scores injected as typed fields; LLM cannot recompute/override them.
Schema validation	Pydantic rejects missing enums/types/invalid evidence references.
Abstention	Unknown/missing evidence is an allowed output.
Audit log	Model, prompt version, evidence IDs, token/cost metadata, retries, and result hash stored.

17.1 Prompt content principle

Prompts should ask for concise reasoning summaries and cited structured conclusions, not hidden chain-of-thought. The stored audit object contains claims, evidence links, counterevidence, and reason codes sufficient to evaluate the result without requiring private reasoning traces.

18. Caching, reproducibility, and cost control

Agent results are expensive relative to deterministic calculations, so every successful node output is cacheable. Historical reruns should reuse identical results when the event, evidence set, prompt, and model configuration are unchanged.

Cache key component	Why it matters
event_id	Identifies the financial decision point.
source_snapshot_id / evidence_hash	Invalidates cache if admissible evidence changes.
retrieval_version	Invalidates if chunk/ranking policy changes.
prompt_version	Separates prompt experiments.
llm_model_id + temperature/config	Separates provider/model behaviour.
agent_schema_version	Separates output contracts.

18.1 Inference settings

- Use low temperature / deterministic mode where supported.
- Cap evidence count and output tokens.
- Retry only deterministic operational failures; do not endlessly resample for a preferred answer.
- Cache final validated outputs for demo mode.
- Record token counts and estimated API cost per event when provider metadata is available.

19. Storage schemas and artefacts

Object	Grain	Required fields
analytics.event_evidence	event x chunk	event_id, chunk_id, rank, leg, relevance_score, source timestamp, retrieval_version
analytics.catalyst_events	event x extracted catalyst	event_id, catalyst_id, type, leg, direction, materiality, relevance, confidence, evidence_ids, model/prompt versions
analytics.agent_findings	event x agent role	event_id, role, thesis, claims JSON, evidence_ids, strength, uncertainty, validation flags, versions
analytics.agent_validation	event	validation_status, completeness, contradiction, invalid claims, break risk, abstain flag, reason codes
analytics.decisions	event x policy version	event_id, final_status, P(revert), break_risk, gate flags, reason_codes, model/policy/source versions
ops.agent_runs	one orchestration run	run_id, event_id, start/end, node statuses, retries, token/cost, cache hits, errors

19.1 Artefacts

- Versioned prompt templates and schema definitions.
- Retrieval configuration and optional embedding artefact if used.
- Cached JSON outputs keyed by event/evidence/model/prompt versions.
- Human-labelled evaluation subset and annotation guide.

20. UI integration

Pair Detail region	Displayed content
Catalyst timeline	Chronological admissible events with affected leg, type, materiality, source timestamp and source link/ID.
Evidence panel	Top source snippets, filing form/accession, section label, acceptance time.
Adversarial analysis	Side-by-side Mean-Reversion and Structural-Break cases with cited claims and acknowledged counterevidence.
Validation	Source completeness, contradiction/invalid-claim flags, break risk, abstention warning.
Final status card	CANDIDATE / REVIEW / NO TRADE / ABSTAIN, calibrated P(revert), break risk, deterministic reason codes, policy version.
Audit expander	Model/prompt/source/retrieval versions, cache status, run timestamp, evidence IDs.

UI principle

The user should be able to move from every agent statement to the underlying stored evidence. The agent prose is not the audit trail; the evidence IDs and deterministic system state are.

21. Agent-specific evaluation hooks

Detailed evaluation belongs in the Backtesting & Evaluation specification, but this pipeline must emit the fields required to measure its contribution.

Metric	Required stored output
Failed-opportunity rejection rate	Final status for every baseline failure.
False-veto rate	Final status for every baseline success.
Event extraction accuracy	Structured catalyst fields on a human-labelled subset.
Evidence validity rate	Claim-to-evidence validation result / manual check.
Abstention rate	ABSTAIN count and reason codes.
Agent disagreement rate	Whether advocates materially conflict.
Source coverage	COMPLETE/PARTIAL/UNKNOWN by event and leg.

21.1 Human-labelled subset

Create a stratified evaluation subset of at least 50 historical events (aim for ~100 if feasible), spanning sectors, successes/failures, and event types. Human reviewers label event_type, affected_leg, materiality, relationship relevance, and whether each selected source genuinely supports the extracted claim.

22. Functions, repository modules, and jobs

Module / function	Responsibility
evidence/select.py::select_admissible_documents	Apply timestamp/form/lookback policy for both legs.
evidence/chunk.py::parse_and_chunk_filing	Convert cached SEC body/exhibits into versioned chunks.
evidence/retrieve.py::rank_event_evidence	Metadata filter + lexical rank + optional semantic rerank.
agents/extract.py::extract_catalysts	Structured event extraction.
agents/advocates.py::run_mean_reversion / run_structural_break	Parallel hypothesis construction.
agents/validate.py::validate_agent_findings	Citations, timestamps, consistency, schema, break risk.
controller/policy.py::apply_decision_policy	Deterministic final status and reason codes.
agents/graph.py::run_agent_pipeline	LangGraph orchestration, retries, cache, persistence.
jobs/run_event_intelligence.py	Batch historical events or current candidates through the pipeline.

22.1 Dependencies

Core dependencies remain Python 3.11, pandas/NumPy, DuckDB/Parquet, Pydantic, LiteLLM, LangGraph, and scikit-learn for deterministic TF-IDF retrieval. An optional local sentence-transformers dependency may be added only if semantic reranking is implemented and versioned.

23. Build sequence and acceptance criteria

Order	Deliverable	Acceptance condition
1	SEC evidence selector	Given historical event, returns only pre-event eligible filings for both legs.
2	Parser/chunker/index	Stored chunk reproduces to raw filing/hash and section metadata.
3	Retrieval	Top evidence pack deterministic for fixed inputs/config.
4	Catalyst extraction	Schema-valid output with valid evidence IDs; NONE/UNKNOWN distinction works.
5	Advocates	Parallel cases use identical evidence and acknowledge counterevidence.
6	Validator	Detects invalid citations/timestamps/contradictions and can recommend ABSTAIN.
7	Controller	Deterministic unit tests cover all CANDIDATE/REVIEW/NO TRADE/ABSTAIN branches.
8	Caching/persistence	Rerun hits cache and reproduces stored output/version lineage.
9	UI	Pair Detail displays citations, cases, validation and final reason codes.
10	Human subset	Extraction/evidence metrics can be computed on labelled events.

- No material agent claim without valid source provenance.
- No historical source timestamp after the event cutoff.
- No LLM-generated market statistics or model probabilities.
- Repeated LLM/tool failure terminates safely.
- Final status is exactly reproducible from persisted validated state + policy version.

Appendix A. Working configuration defaults

Parameter	Working default
Canonical text source	SEC EDGAR
Recent event lookback	8-Ks accepted in prior 30 calendar days
Periodic context	Latest 10-Q/10-K <=180 days; extend <=400 days if required
News	Optional current/demo GDELT enrichment; excluded from headline historical Model B and Full-System evaluation
Chunk size	~800-1,200 words, ~100-word overlap when splitting
Evidence pack	Max ~12 chunks, normally <=6 per leg
Core ranker	Metadata filtering + TF-IDF/cosine ranking
Semantic rerank	Optional local embeddings only
LLM abstraction	LiteLLM; configured model must support reliable structured output
Orchestration	LangGraph typed state
Temperature	Low/deterministic where supported
Candidate P(revert)	>=0.65 starting threshold
Review probability band	0.45-0.65 starting band
High break risk	Hard NO TRADE
Unknown/failed evidence	ABSTAIN when classification is not defensible

Appendix B. JSON-style field reference

B.1 Catalyst object

Field	Type
catalyst_id	string
event_type	enum
affected_leg	enum
direction	enum
materiality	enum
relationship_relevance	enum
confidence	float 0-1
evidence_ids	list[string]
source_timestamp	datetime
summary	string

B.2 Advocate object

Field	Type
role	enum
thesis	string
supporting_claims	list[{claim,evidence_ids}]
counterevidence_acknowledged	list[string]
evidence_strength	LOW/MEDIUM/HIGH
uncertainties	list[string]
source_ids	list[string]

B.3 Controller output

Field	Type
event_id	string
final_status	CANDIDATE/REVIEW/NO TRADE/ABSTAIN
reason_codes	list[enum]
calibrated_probability	float
break_risk	LOW/MEDIUM/HIGH/UNKNOWN
policy_version	string
model_version	string
source_snapshot_id	string
decision_timestamp	datetime

References and source basis

Source	Use
Complete Project Overview	Sections 9-11: event intelligence, agent roles, orchestration, decision controller and risk gates.
Data Architecture & Source Strategy	Exact SEC EDGAR retrieval, timestamp, raw/curated storage, document-chunk schema, optional GDELT policy.
Supervised ML Pipeline	Event identity, quantitative state, calibrated P(revert), SHAP, version lineage.
FINS5557 Week 3 materials	Financial NLP, embeddings/retrieval, structured classification context.
FINS5557 Week 4 materials	Agent perception-reasoning-action, multi-agent/adversarial trading architecture and structured communication.
FINS5557 Week 5 materials	Hallucination, prompt injection, model/vendor risk and governance controls.